Understanding Automation Worksheet

Name(s
Date:

Instructions

Work with your partner to analyze the scenario assigned to you. Answer the questions clearly and be prepared to share your thinking.

Scenario Description:

Scenario A

Condition.

A dorm hallway has lights that do not need to stay on all day. When a student walks into the hallway at night, the lights come on without anyone touching a switch. After the hallway is empty for a few minutes, the lights turn off again.

CONTROL ACTION

RESPONSE

Scenario B

At the entrance of a campus building, students do not need to push the door open. As a person gets close to the entrance, the door opens by itself and then closes again after the person passes through.

Scenario C

At the entrance to a campus gym, students do not need to manually unlock the door. When a student taps their ID card on a scanner, the system recognizes the card and the door unlocks automatically, allowing access to the facility.

Scenario D

In a public restroom, students do not have to turn a faucet handle. When hands are placed under the faucet, water begins to flow, and when the hands move away, the water stops.

Understanding the System

Scenario #: _____

1. **What condition is being detected?**

a) Student Presence at night.
b) Person get closer or faculty from door
c) Proximity of id to the scanner
d) Proximity of hands to faucet

2. **What action happens automatically?**

a) Light turns on an off
b) Door opens & closes
c) Door unlocks
d) faucet turns on & off

3. **What result does the system produce?**

a) Hallway lights up.
b) Allow access ☺
c) Access to the gym
d) Student can wash hands.

4. **Can you think of another example that works in a similar way?**

Pet feeder starts at a specific time & a certain amount of food is dispensed